Functional Dependency 🔑

A **functional dependency** is a relationship that exists when one set of attributes in a table uniquely determines another set of attributes.[1] It is a fundamental concept in relational database theory and is crucial for database normalization.[2] Functional dependencies help us understand the relationships between attributes and identify potential data anomalies.[3]

A functional dependency is denoted as **`X → Y`**, which means that `X` **functionally determines** `Y`. [4]Here, `X` is the **determinant** and `Y` is the **dependent**.[5] This relationship holds true if, for any two tuples (rows) in the table, whenever the values for the `X` attributes are the same, the values for the `Y` attributes must also be the same.[6]

- **Example**: In a `Student` table with attributes `(StudentID, StudentName, City)`, if each `StudentID` is unique, then `StudentID` functionally determines `StudentName` and `City`. [7]So, we can write `StudentID → StudentName` and `StudentID → City`.[8]

Properties of Functional Dependency

The properties of functional dependencies, also known as **Armstrong's Axioms**, are a set of rules that allow us to infer all possible functional dependencies from a given set.[9] These axioms are sound (they don't generate invalid dependencies) and complete (they can generate all valid dependencies).[10]

1. Reflexivity (Trivial Dependencies)

If `Y` is a subset of `X`, then `X` functionally determines `Y`. [11]This is a trivial dependency because it is always true and doesn't provide any new information about the data.[12]

- **Axiom**: If `Y ⊆ X`, then `X → Y`.
- **Example**: If you have a set of attributes `{StudentID, StudentName}`, then `{StudentID, StudentName}` will always determine `{StudentID}`. So, `{StudentID, StudentName} → StudentID`.

2. Augmentation

If `X` functionally determines `Y`, and `Z` is another set of attributes, then `X` along with `Z` will also determine `Y` along with `Z`. [13]This means we can add attributes to both sides of the dependency.

- **Axiom**: If `X → Y`, then `X Z → Y Z`.
- **Example**: If `StudentID → StudentName`, then `{StudentID, Age} → {StudentName, Age}`.

3. Transitivity

If `X` functionally determines `Y`, and `Y` functionally determines `Z`, then `X` must also functionally determine `Z`. [14]This is the basis for third normal form (3NF).

- **Axiom**: If `X → Y` and `Y → Z`, then `X → Z`.

- **Example**: If `StudentID → CourseID` and `CourseID → CourseTitle`, then `StudentID → CourseTitle`.